

6-Axis Robotic Arm Kinematics & Path Planning

Abstract

This project focuses on designing a 6-axis robotic arm using ROS2, MoveIt, and RViz with inverse kinematics and collision-free motion planning.

Objective

- Design 6-DOF robotic arm
- Apply inverse kinematics
- Generate smooth trajectories
- Collision-free motion planning

Technologies Used

Python, ROS2, MoveIt, RViz, URDF/Xacro, YAML, Kinematics

System Flow

User Input → IK Solver → Joint Angles → MoveIt Planner → Trajectory → Collision Check → Execution → RViz

Working Principle

Robot is modeled in URDF. Target position is given. IK computes joint angles. MoveIt generates safe trajectory and executes motion.

Kinematics

Forward Kinematics: $T = A1 \times A2 \times A3 \times A4 \times A5 \times A6$

Inverse Kinematics: $(X,Y,Z) \rightarrow \theta1-\theta6$

Applications

Industrial automation, Pick and place, Medical robotics, Warehouse automation

Conclusion

The project successfully demonstrates robotic motion planning using ROS2 and MoveIt with collision-free trajectory execution.